MRUTUNJAY YADAV

MERN STACK DEVELOPER

mrutunjay765@gmail.com

H504, Rashmi Classic , Link Road, Vasai East-401209

github.com/Mrutunjay113

portfolio-2-0-alpha.vercel.app

8286474981

01 Apr 2001

linkedin.com/in/mrutunjay-yadav-48a856168

Profile

MERN Stack Developer with 3+ years of experience building scalable, high-performance web applications using Next.js, React.js, Node.js, Express.js, TypeScript, and MongoDB. Design responsive, performance-optimized user interfaces and secure RESTful APIs, with a strong focus on MongoDB data modeling, authentication/RBAC, and server-state management. Own features end-to-end through production releases on Vercel, delivering measurable improvements in page load, API latency, and data consistency.

Professional Experience

FilamentAi Pvt. Ltd. Aug 2023 – present | MUMBAI, India

MERN Stack Developer

- Started as a Frontend Developer and, after the first 3 months, was transitioned into a MERN Stack Developer role — owning work end-to-end from responsive UI in Next.js and React.js to REST APIs in Express.js, MongoDB data modeling, and production releases on Vercel.
- Designed, documented, and shipped secure REST APIs for authentication, user management& reporting, following consistent conventions for validation, error handling, and versioned routing.
- Designed MongoDB schemas, implemented compound indexes, and optimized aggregation pipelines to support efficient data retrieval and reporting across analytics-driven applications.
- Improved application performance using Next.js Server Components, code splitting, lazy loading, and optimized rendering strategies.
- Implemented authentication and authorization workflows using Better Auth, including session management and role-based access control (RBAC).
- Developed reusable, modular UI components using Next.js, React.js, and Tailwind CSS to maintain consistency and simplify feature development across product modules.
- Collaborated with AI engineering teams to integrate computer-vision event data into MongoDB-backed applications, supporting CCTV analytics across safety violations, fire and smoke detection, PPE compliance, attendance, and other monitoring use cases.
- Managed application configuration, environment variables, and production deployments using Vercel.

Projects

Attendance & Productivity Analytics Platform

- Developed a full-stack workplace attendance and productivity analytics platform that processes camera-based desk occupancy timelines to generate attendance summaries, productive hours, break durations, overtime metrics, and department-level analytics
- Developed an admin dashboard with interactive charts, trend views, and date-range and department filters, so managers can compare productivity across teams and time periods.
- Integrated TanStack Query for client-side server-state management and caching, alongside path revalidation after mutations to keep dashboard data consistent with database updates.
- Optimized MongoDB data access using Mongoose lean queries, selective field projections, compound indexes, aggregation pipelines, and parallel database operations to improve query efficiency and reduce unnecessary data processing.
- **Tech Stack:** Next.js ,React , MongoDB, Mongoose, NextAuth.js, TanStack Query, Tailwind CSS, Zod

Campus Occupancy & Identity Platform

- Developed a full-stack campus monitoring platform with real-time occupancy tracking across buildings, floors, and locations, including assembly-point headcounts and historical analytics.
- Implemented identity enrollment and face-event review workflows, including invite-based registration, photo uploads, approval processes, and audit trails for review decisions.
- Implemented REST APIs using Next.js Route Handlers for occupancy analytics, spatial management, identity operations, attendance, and visitor workflows, integrating external face-recognition and media services.
- Optimized MongoDB analytics using compound indexes, aggregation pipelines, parallel queries, and daily pre-aggregations to support efficient historical reporting and face-event filtering.
- Integrated Socket.IO-based real-time updates with Server-Sent Events (SSE) and REST snapshot fallbacks to support live dashboard monitoring.
- Implemented Better Auth authentication, role-based administration & Zod validation.
- **Tech Stack:** Next.js, React, MongoDB, Mongoose, Better Auth, Socket.IO, TanStack Query, Zod, Recharts, Tailwind CSS

Digital Market Place

- Buy & Download Digital Products *Self-driven product.*
- Built an end-to-end marketplace where sellers list digital products and buyers pay, then receive secure, time-limited download links — covering catalog browsing, cart, checkout, order fulfillment and post-purchase download.
- Designed secure Stripe checkout with webhook-driven order state, and protected download endpoints with signed URLs so only paying customers can retrieve product files.
- Used Payload CMS as the headless admin to manage products, categories, orders and users without a separate CMS UI; exposed typed tRPC procedures from Payload so the Next.js frontend stays fully type-safe end-to-end.
- Modeled MongoDB collections for products, orders, downloads and audit logs, and built admin dashboards in Shadcn UI + Tailwind CSS for revenue, top-selling products and download counts
- **Tech Stack:** Next.js, TypeScript, tRPC, Stripe, Payload CMS, Shadcn UI, MongoDB, Tailwind CSS

PPE & Workplace Safety Analytics Dashboard

- Developed a computer vision-powered workplace safety dashboard to monitor PPE compliance and industrial safety events across multiple sites.
- Built a safety event data pipeline to store and retrieve camera-based detections, including missing helmet and other PPE violations.
- Created interactive analytics dashboards with KPI summaries, charts, and historical trends to track safety incidents and compliance.
- Implemented date, use-case, and client-based filtering for investigating safety events across different sites and time periods.
- Developed Next.js API route handlers for detection records, aggregated counts, and media-related resources.
- Designed structured MongoDB/Mongoose data models to support real-time and batch-based updates from edge computer vision systems.

Tech Stack: Next.js · React.js · Node.js · MongoDB · Mongoose · Next.js App Router · HeroUI

Skills

Programing Language

- Javascript
- java

Backend Technologies

- Node.js
- Express.js

Database

- MongoDB
- MySql

Libraries & Tools

- TanStack Query
- TanStack Form
- Better Auth

Frontend Technologies

- Next.js
- React.js
- Tailwind CSS
- HTML5
- CSS3

Education

MCA
Thakur Institute of Management Studies Career Development& Research
- Completed in 2025 with **8.19 CGPI**.

BSc Cs
M.V.L.U
8.5 CGPA

Languages

English

Hindi

Interests

- Playing Games
- Playing Cricket